Tier 1 Tier 2 Tier 3 Tier 4

(a) r = 0.85

RDF error [%]

MatterSim-v1.0.0-5M
UMA-M-P1
UMA-S-P1
MACE-MP-0
orb-v3-direct-20-omat
orb-v2
orb-v3-conservative-inf-omat
NequIP-OAM-L
PET-OMAT-XL
MACE-MPA-0
eSEN-30M-OAM
MACE-MH-1-OMAT
PET-OAM-XL
EquiformerV2
GRACE-2L-OAM

(b) r = -0.64

MACE-MP-0
MatterSim-v1.0.0-5M
orb-v2
NequIP-OAM-L
MACE-MPA-0
PET-OAM-XL
EquiformerV2
eSEN-30M-OAM
GRACE-2L-OAM

(c) r = 0.33

MatterSim-v1.0.0-5M
MACE-MP-0
orb-v2
NequIP-OAM-L
MACE-MPA-0
PET-OAM-XL
EquiformerV2
eSEN-30M-OAM
GRACE-2L-OAM

(d) r = 0.54

Pressure histogram error [%]

MACE-MPA-0
MACE-MP-0
orb-v3-direct-20-omat
EquiformerV2
MatterSim-v1.0.0-5M
orb-v3-conservative-inf-omat
orb-v2
NequIP-OAM-L
PET-OAM-XL
PET-OMAT-XL
UMA-S-P1
GRACE-2L-OAM
UMA-M-P1
eSEN-30M-OAM
MACE-MH-1-OMAT

(e) r = -0.61

MACE-MP-0
MACE-MPA-0
MatterSim-v1.0.0-5M
EquiformerV2
orb-v2
PET-OAM-XL
NequIP-OAM-L
GRACE-2L-OAM
eSEN-30M-OAM

(f) r = 0.41

MACE-MPA-0
MACE-MP-0
MatterSim-v1.0.0-5M
EquiformerV2
orb-v2
PET-OAM-XL
NequIP-OAM-L
GRACE-2L-OAM
eSEN-30M-OAM

(g) r = 0.73

VDOS error [%]

MACE-MP-0
orb-v2
orb-v3-direct-20-omat
MatterSim-v1.0.0-5M
MACE-MPA-0
NequIP-OAM-L
UMA-S-P1
UMA-M-P1
orb-v3-conservative-inf-omat
MACE-MH-1-OMAT
PET-OMAT-XL
PET-OAM-XL
eSEN-30M-OAM
GRACE-2L-OAM
EquiformerV2

Force RMSE [eV/Å]

(h) r = -0.75

MACE-MP-0
orb-v2
MatterSim-v1.0.0-5M
MACE-MPA-0
NequIP-OAM-L
eSEN-30M-OAM
PET-OAM-XL
GRACE-2L-OAM
EquiformerV2

F1 score

(i) r = 0.20

MACE-MP-0
orb-v2
MatterSim-v1.0.0-5M
MACE-MPA-0
NequIP-OAM-L
PET-OAM-XL
GRACE-2L-OAM
eSEN-30M-OAM
EquiformerV2

κ_{SRME} score